

Second Order Partial Derivatives

For problems 1 – 3 verify Clairaut's Theorem for the g.....

1. $Q(s, t) = \ln(st) - s^4 \sin(6t) + st$

2. $f(u, w) = \sin(uw) + 4u^2 w^{-2}$

3. $f(x, y) = e^{xy} \sin(y)$

For problems 4 – 9 find all 2nd order derivatives for the given function.

4. $h(x, y) = x^4 y^{-2} - 4xy + e^{7y} + \ln(2x)$

5. $A(u, v) = u^2 \cos(3v) + \ln\left(\frac{u}{4v^2}\right)$

6. $g(v, w) = \ln(v \sin(w)) + \sin(v \ln(w))$

7. $f(x, y) = \cos(x^2 + y^2) - \sin(xy)$

8. $h(x, y, z) = 7x^3 y^2 z^4 + 8$

9. $Q(u, v, w) = u^4 \sin(w^2) - \frac{2v}{u^4} + \ln(v^2 w)$

